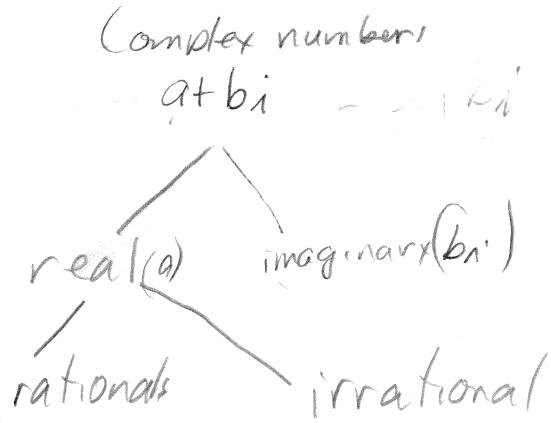
Lecture 15: Section 2.4
Complex Numbers



Imaginary unit

Complex numbers

Operations with complex numbers

Complex conjugate

Rationalize the denominator

Square roots of negative numbers

Complex solutions of quadratic equations

Consider the equation $x^2 = -1$.

<u>Def.</u> The **imaginary unit**, i, is the number such that

$$* \quad i^2 = -1 \quad \text{or} \quad i = \sqrt{-1} \quad *$$

Power of i

$i^1 = \sqrt{-1} = i \qquad i^2 = -1 \qquad \boxed{i^3 = -i} \qquad \boxed{i^4 = 1}$

$i^2 \cdot i$, $-1 \cdot i$ $\qquad$ $i^2 \cdot i^2$, $-1 \cdot -1$

$\boxed{i^5 = i}$ $\quad$ $i^4 \cdot i$, $1 \cdot i$ $\qquad$ $\boxed{i^6 = -1}$ $\quad$ $i^4 \cdot i^2$, $1 \cdot -1$ $\qquad$ $\boxed{i^7 = -i}$ $\quad$ $i^4 \cdot i^3$, $1 \cdot (-i)$ $\qquad$ $\boxed{i^8 = 1}$ $\quad$ $i^4 \cdot i^4$, $1 \cdot 1$

Therefore, every integer power of i can be written as $i,\ -1,\ -i,\ 1$.

In general, divide the exponent by 4 and rewrite:

<u>ex.</u> 1) $\boxed{i^{85} = i}$ $\qquad$ 2) $\boxed{(-i)^{85} = -i}$

$= (i^4)^{21} i = 1^{21} \cdot i = 1 \cdot i =$

$= (-1 \cdot i)^{85} = (-1)^{85} \cdot i^{85} = -1 \cdot i = \boxed{-i}$

$(i^4)^{25} = 1^{25} = 1$ $\qquad$ 3) $\boxed{i^{100} = 1}$ $\qquad$ 4) $(-i)^{-18} = \dfrac{1}{(-i)^{18}} = \dfrac{1}{(-1 \cdot i)^{18}} =$

Checkpoint: Lecture 15, problem 1

$= \dfrac{1}{(-1)^{18} \cdot i^{18}} = \dfrac{1}{1 \cdot (i^4)^4 i^2}$

$= \dfrac{1}{1 \cdot 1^4 \cdot (-1)} = \dfrac{1}{-1} = \boxed{-1}$

Def. **Complex numbers** are numbers of the form $a + bi$, where a and b are real numbers.

a is the **real part** and b is the **imaginary part** of the complex number $a + bi$.

$a + bi$ is called the **standard form** of a complex number.

ex. Write the number -5 as a complex number in standard form.

complex #s $a+bi$

real (a) imag. (bi)

rat irr

$-5 = \boxed{-5 + 0i}$

NOTE: The set of real numbers is a subset of the set of complex numbers. ~~$a+bi$~~

$6+0i$ If $b = 0$, the number $a + 0i = a$ is a real number. 6

$0+7i$ If $a = 0$, the number $0 + bi = bi$, is called a **pure imaginary number**. $7i$

Equality of Complex Numbers

$a + bi = c + di$ if and only if $a = c$ and $b = d$

$6 + 3i = 6 + 3i$

Operations with Complex Numbers

Sum: $(a + bi) + (c + di) = (a+c) + (b+d)i$

$= a + bi - c - di =$

Difference: $(a + bi) - (c + di) = (a-c) + (b-d)i$

FOIL **Multiplication:** $(a + bi)(c + di) = ac + adi + bci + bdi^2$ $= -1$

$= ac + adi + bci - bd = (ac - bd) + (ad + bc)i$

NOTE: Use the distributive property (FOIL) and remember that $i^2 = -1$.

ex. Write in standard form:

1) $3(2 - 5i) - (4 - 6i) = 6 - 15i - 4 + 6i =$

$= \boxed{2 - 9i}$

2) $(2 + 3i)(4 + 5i) = 8 + 10i + 12i + 15i^2$ $= -1$

$= 8 + 22i - 15 = \boxed{-7 + 22i}$

Checkpoint: Lecture 15, problem 2

$z = a + bi$

$\bar{z} = a - bi$

Complex Conjugates Used for simplifying.

Def. The **conjugate (complex conjugate)** of a complex number $a + bi$ is $a - bi$

ex. Find the conjugate of the following:

1) $-1 - 3i$ $\quad -1 + 3i$

$0 - 2i \to 0 + 2i$

2) $-2i$ $\quad 2i$

3) $5 + 0i$ $\quad 5 - 0i = 5$

Checkpoint: Lecture 15, problem 3

NOTE: A real number is its own conjugate.

FOIL

Theorem: The product of a complex number and its conjugate is a nonnegative real number:

$$(a+bi)(a-bi) = a^2 - \cancel{abi} + \cancel{abi} - b^2 i^2 \quad i^2 = -1$$

$$= a^2 - b^2(-1)$$

$$= a^2 + b^2$$

Rationalize the denominator - no i in the bottom

To express a fraction with a complex number in the denominator in standard form, multiply the numerator and denominator by the conjugate of the denominator.

ex. Rationalize and write in standard form.

$$\left(\frac{1}{3+2i}\right)\left(\frac{3-2i}{3-2i}\right) = \frac{3-2i}{3^2+2^2} = \frac{3-2i}{13} =$$

$$= \frac{3}{13} + \frac{-2i}{13} = \frac{3}{13} + \left(\frac{-2}{13}i\right) = \boxed{\frac{3}{13} - \frac{2}{13}i}$$

ex. Perform the operation $\dfrac{1-3i}{4+2i}$ and write in standard form.

$$\left(\frac{1-3i}{4+2i}\right)\left(\frac{4-2i}{4-2i}\right) = \frac{4-2i-12i+6i^2}{4^2+2^2} = \frac{4-14i-6}{20} = \left(\frac{-2-14i}{20}\right)\left(\frac{\frac{1}{2}}{\frac{1}{2}}\right)$$

$$= \frac{-1-7i}{10} = \boxed{\frac{-1}{10} - \frac{7}{10}i}$$

Checkpoint: Lecture 15, problem 4

Square Roots of Negative Numbers

Def. If a is a positive real number then the **principal square root** of $-a$ is given by

$$\sqrt{-a} = \sqrt{-1 \cdot a} = \underbrace{\sqrt{-1}}_{=i} \cdot \sqrt{a} = i\sqrt{a} = \sqrt{a}\,i = \sqrt{a}\,i$$

ex. Evaluate:

1) $\sqrt{-64} = \sqrt{-1 \cdot 64} = \sqrt{-1} \cdot \sqrt{64} = i\,8 = \boxed{8i}$

short-cut $\sqrt{64}\,i = 8i$

2) $\sqrt{-200} = \sqrt{-1 \cdot 200} = \sqrt{-1} \cdot \sqrt{\overset{10^2 \cdot 2}{200}} = i \cdot 10\sqrt{2}$

$\boxed{= 10i\sqrt{2} = 10\sqrt{2}\,i}$

short-cut $= \sqrt{200}\,i = 10\sqrt{2}\,i$

NOTE: Convert the negative radicand of a square root to a complex number before multiplying.

($\sqrt{a}\sqrt{b} = \sqrt{ab}$ is NOT valid if both a and b are negative numbers)

ex. 1) $\sqrt{-3}\sqrt{-3} = \sqrt{3}\,i \cdot \sqrt{3}\,i = 3i^2 = \boxed{-3}$

2) $\sqrt{(-3)(-3)} = \sqrt{9} = \boxed{3}$

ex. Perform the operations and simplify:

1) $\sqrt{-8}+\sqrt{-18} = \sqrt{8}\,i + \sqrt{18}\,i$

$= 2\sqrt{2}\,i + 3\sqrt{2}\,i = \sqrt{2}\,i\,(2+3)$

$= \boxed{5\sqrt{2}\,i}$ $\qquad = \sqrt{2}\,i\,(5)$

2) $\sqrt{-5}\sqrt{-15} = \sqrt{5}\cdot i\cdot\sqrt{15}\,i = \sqrt{\overset{5\cdot 3}{75}}\,i^2 =$

$5\sqrt{3}\,(-1) = \boxed{-5\sqrt{3}}$

long way $(1+\sqrt{2}\,i)^2 = (1+\sqrt{2}\,i)(1+\sqrt{2}\,i)$ FOIL

3) $(1+\sqrt{-2})^2 = (1+\sqrt{2}\,i)^2 = 1^2 + 2(1)(\sqrt{2}\,i) + (\sqrt{2}\,i)$

$= 1 + 2\sqrt{2}\,i + 2\,i^2$ $\quad i^2 = -1$

$= 1 + 2\sqrt{2}\,i - 2$

$= \boxed{-1 + 2\sqrt{2}\,i}$

Checkpoint: Lecture 15, problem 5

Complex Solutions of Quadratic Equations

In the complex number system, the solutions of the quadratic equation $ax^2+bx+c=0$, where a, b and c are real numbers and $a \neq 0$ are given by the formula

$$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$$

$a=1 \quad b=4 \quad c=10$

ex. Solve the equation $x^2 + 4x + 10 = 0$ in the complex number system.

$$x = \frac{-4 \pm \sqrt{4^2-4(1)(10)}}{2(1)} = \frac{-4 \pm \sqrt{-24}}{2} = \frac{-4 \pm \sqrt{24}\,i}{2} =$$

$$= \left(\frac{-4 \pm 2\sqrt{6}\,i}{2}\right)\left(\frac{\frac{1}{2}}{\frac{1}{2}}\right) = \frac{-2 \pm \sqrt{6}\,i}{1} = -2 \pm \sqrt{6}\,i = x$$

or $x = -2 + \sqrt{6}\,i$
$x = -2 - \sqrt{6}\,i$

NOTE: The complex solutions are always conjugates of each other!

$$-2 + \sqrt{6}\,i + (-2 - \sqrt{6}\,i)$$
$$-2 + \cancel{\sqrt{6}\,i} - 2 - \cancel{\sqrt{6}\,i} = -4$$

To summarize:

For the equation $ax^2 + bx + c = 0$ with real coefficients, find the discriminant $D = b^2 - 4ac$.

1. If $D > 0$, the equation has two unequal real number solutions.

2. If $D = 0$, the equation has a repeated real solution.

3. If $D < 0$, the equation has two complex conjugate solutions that are not real numbers.

ex. Determine the character of the solution(s) of the following equation:

1) $2x(6 - x) = 18$

$12x - 2x^2 = 18$

$\frac{1}{2}[2x^2 - 12x + 18 = 0]$

$x^2 - 6x + 9 = 0$

$D = b^2 - 4ac$

$D = 36 - 4(1)(9) = 0$

one repeated real sol.

2) $x^{-2} - 3x^{-1} + 3 = 0$

$x^2\left[\frac{1}{x^2} - \frac{3}{x} + 3 = 0\right] \rightarrow 1 - 3x + 3x^2 = 0$

$3x^2 - 3x + 1 = 0$

$D = 9 - 4(3)(1) = -3$

two complex conjugate solutions